Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2026

Course: B.Tech | *Time: 03 Hours*
Code: 155603 | *Subject: Ad hoc & Sensor Networks* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven questions only):*** **[2 x 7 = 14]**

(a) A WSN application requires each sensor to remain asleep most of the time and wake quickly when traffic is detected. Which MAC design objective is most directly being addressed?
(i) Maximum route length
(ii) Energy conservation through reduced idle listening
(iii) Elimination of synchronization
(iv) Maximum transmit power

(b) In pure ALOHA, if the normalized offered load is G=0.5, the throughput $S=Ge^{(-2G)}$ is approximately
(i) 0.092
(ii) 0.184
(iii) 0.303
(iv) 0.500

(c) Which statement best distinguishes X-MAC from traditional S-MAC operation?
(i) X-MAC uses short strobed preambles to reduce overhearing and waiting
(ii) X-MAC requires GPS for every transmission
(iii) X-MAC is a proactive routing protocol
(iv) X-MAC eliminates contention completely

(d) AODV maintains loop freedom primarily through the use of
(i) Cluster-head rotation
(ii) Destination sequence numbers
(iii) Geographic coordinates only
(iv) Fixed TDMA slots

(e) In SPIN, unnecessary transmission of complete sensor data is reduced mainly by
(i) Negotiating data using metadata before transmission
(ii) Maintaining complete routes to every node
(iii) Using GPS coordinates for every packet
(iv) Disabling data aggregation

(f) A node has a 12 kJ battery and consumes an average of 20 mW. Ignoring all losses, its approximate operating time is
(i) 60 s
(ii) 600 s
(iii) 6.94 days
(iv) 69.4 days

(g) If three anchors used by a centroid localization algorithm are at (0,0), (9,0) and (0,12), the estimated position is
(i) (0,0)
(ii) (4,3)
(iii) (9,12)
(iv) (3,4)

(h) Which statement about TDOA localization is most appropriate?
(i) It estimates position from differences in signal arrival times
(ii) It requires no timing-related measurements
(iii) It is identical to RSSI localization
(iv) It determines position only from packet sequence numbers

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(i) A wormhole attack is especially dangerous to routing because colluding attackers can

(i) Create an apparently short tunnel between distant network locations
(ii) Only listen passively without affecting routes
(iii) Change the physical battery capacity of nodes
(iv) Prevent all nodes from generating sensor data

(j) In a VANET, very rapid topology changes most directly increase the difficulty of

(i) Performing arithmetic at a sensor
(ii) Measuring battery voltage
(iii) Storing a fixed node identifier
(iv) Maintaining valid multi-hop routes

Q.2 (a) Explain the major design issues of wireless sensor networks, including scalability, fault tolerance, limited energy and constrained bandwidth, and discuss how these requirements influence network architecture. [7]

(b) Explain layered and cross-layer design approaches in ad hoc and sensor networks and discuss the advantages and possible disadvantages of sharing information across protocol layers. [7]

Q.3 (a) Explain the operation of S-MAC, T-MAC, B-MAC and X-MAC with emphasis on idle listening, overhearing and duty cycling. [7]

(b) Explain contention-based and scheduling-based MAC protocols and compare their suitability under light and heavy network traffic. [7]

Q.4 (a) Explain route discovery and route maintenance in DSR and compare source routing in DSR with next-hop routing in AODV. [7]

(b) Explain the principles of Directed Diffusion and GEAR and discuss how data-centric and geographic/energy-aware routing differ from conventional address-centric routing. [7]

Q.5 (a) Explain data aggregation and clustering in WSN and discuss their role in reducing redundant transmissions and conserving energy. [7]

(b) Explain the purpose of reliable transport in sensor networks and distinguish the basic objectives of RMST, PSFQ and ESRT. [7]

Q.6 (a) Explain GPS-based, range-based and range-free localization and discuss the limitations of GPS for large-scale low-power sensor deployments. [7]

(b) Explain the APIT localization technique and contrast its basic idea with Centroid and DV-Hop localization. [7]

Q.7 (a) Explain the need for data dissemination and query processing in WSN and discuss the trade-off between continuous reporting and query-driven reporting. [7]

(b) Explain the concept of mobile sink-based data collection and discuss how sink mobility can mitigate the energy-hole problem near a static sink. [7]

Q.8 (a) Explain intrusion detection and privacy-preserving techniques in wireless sensor networks and discuss how they complement cryptographic secure routing. [7]

(b) Explain the networking characteristics of MANETs, VANETs and cognitive radio sensor networks and discuss how AI/ML can support prediction or adaptive decision-making in these systems. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Adaptive duty cycling and asynchronous preamble-sampling MAC in WSN
(b) Data-centric and energy-aware routing in WSN
(c) Mobile-sink data collection, energy-hole mitigation and query processing
(d) Trust, privacy and intelligent security in MANET/VANET/CRSN

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